Ronak Gupta

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Education

Vivekananda Institute of Professional Studies — **2024 – Present**
B.Tech in Computer Science and Engineering (CSE) — *Delhi, India*

- **CGPA:** 8.80

Certifications

Elite NPTEL Certificate: Mathematical Foundations for ML | *IISc Bangalore (Silver)* — Monsoon 2025
NPTEL Certificate: Getting Started with Competitive Programming | *IIT Gandhinagar* — Spring 2025
Summer Training: Cyber Security and AI/ML | *IIT Roorkee* — Jun 2025 – Jul 2025

Achievements

Codeforces: Pupil (1200+)
Indian Olympiad Qualifier in Mathematics (IOQM) — 2023

Projects

SchemaForge – *Autonomous Database Migration & Refactoring Agent* | *Python, AST, MCP* — **Link**

- Engineered an autonomous migration agent parsing codebase ASTs and introspecting PostgreSQL schema to construct deterministic **impact graphs** (tables → ORM models → API endpoints) with Mermaid visualization.
- Architected **two-phase expand/contract** migration workflows with dual-write models, automated data backfills, and contract-gate validation to guarantee zero downtime under live traffic.
- Built isolated verification inside a **Daytona sandbox** validating Alembic migrations, data parity, and query benchmarks before gated production execution via custom **MCP servers**.

MiniRelDB – Custom Relational Database Engine | *Java, B+ Trees, WAL* — **Link**

- Built a relational database engine from scratch, implementing SQL operations, query parsing, and execution.
- Designed a **page-based storage** engine with heap files, **buffer management**, **B+ Tree indexing**, and **Bloom Filters** for efficient data access.
- Implemented **Write-Ahead Logging (WAL)**, **crash recovery**, and **Hash Join optimization**, supporting persistent storage and fast lookups on 10K+ records.

Random Number Generators: Security Analysis | *Python, C/C++, BLAKE3* — **Link**

- Analyzed **RNG implementations** across C, C++, Python, and Java, studying vulnerabilities including state recovery, predictability attacks, weak seeding, and cryptographic misuse.
- Designed and implemented B3CTR and B3GUARD, two **BLAKE3-based** RNG architectures incorporating entropy mixing, counter-based generation, reseeding, and bias-free sampling techniques.
- Evaluated randomness quality using the **NIST SP 800-22 statistical test suite**, achieving **17/17 test passes** and comparing results against traditional PRNG approaches.

TuxPages – *Technical Blog Web Application* — **Link**

- Co-developed a technical research journal with publishing, authentication, and content management features, attracting **50+ registered users** and hosting **20+ published blogs**.
- Configured secure PostgreSQL connectivity using environment variables, authentication, SSL, and implemented SMTP email delivery with TLS and App Passwords for account verification and notifications.
- Resolved npm package dependency conflicts and maintained consistent lockfiles to ensure reproducible builds across development environments.

Games Using SDL2 | *C, SDL2, GCC* — **Link**

- Developed multiple games using the **SDL2 library**, implementing graphics rendering, event handling, and real-time user interaction.
- Built games such as **Tic-Tac-Toe** and **2D Platformer** with proper memory management and modular C design.

Technical Skills

Programming Languages: C/C++, Python, Java, SQL (PostgreSQL, MySQL), Bash, Assembly Language, PHP
Libraries & Frameworks: FastAPI, SQLAlchemy, Alembic, SDL2, NumPy, Pandas, Scikit-learn, Matplotlib
Tools & Environments: Git, Docker, Linux (Arch), VS Code, Vim, Arduino, LaTeX, Jupyter Notebook
Soft Skills: Touch Typing

Positions of Responsibility

ADVAIT VSET (Artificial Intelligence Club) | *Technical Lead* — **Feb 2025 – Present**

- Led technical initiatives and conducted AI learning sessions, guiding members in AI and ML projects.

Srijan (SDG Club) | *Corporate Affairs and PR* — **Nov 2024 – Present**

- Coordinated outreach with external organizations to support sustainability and social impact initiatives.

Freedom Employability Academy — **2019 – 2022**
Mentor — *Delhi, India*

- Supported a cohort of 20 students during training sessions by facilitating discussions and learning activities.
- Assisted in planning and delivering interactive learning sessions while supporting student engagement.